

Taiwan Cancer Registry

Temporal Trends in Cancer Incidence 2003–2020

Overview

Using first-primary cancer records for 78,619 patients registered 2003–2020, we computed annual site-specific fractions and tested monotone trend (Spearman ρ). p -values are autocorrelation-corrected (Chelton 1983; lag-1 AR removed from residuals; effective n computed as $n*(1-\phi)/(1+\phi)$).

Hypothesis Results

- H1 C22 liver HCC declining $\rightarrow$ CONFIRMED
 $\rho=-0.983$, $p_{\text{corr}}<0.001$ ($\phi=0.43$, $n_{\text{eff}}=7$)
- H2 UADT betel sites declining post-2006 $\rightarrow$ SPLIT
C13 hypopharynx: $\rho=-0.773$, $p_{\text{corr}}<0.001$ (falling)
C12 pyriform: $\rho=+0.562$, $p_{\text{corr}}=0.0042$ (rising)
- H3 Hormonal sites rising (metabolic syndrome) $\rightarrow$ PARTIAL
C54 endometrial: $\rho=+0.925$, $p_{\text{corr}}=0.0085$ (confirmed)
C50 breast: $\rho=-0.238$, $p=0.3408$ (ns, $\phi=0.77$)

Bonus Findings

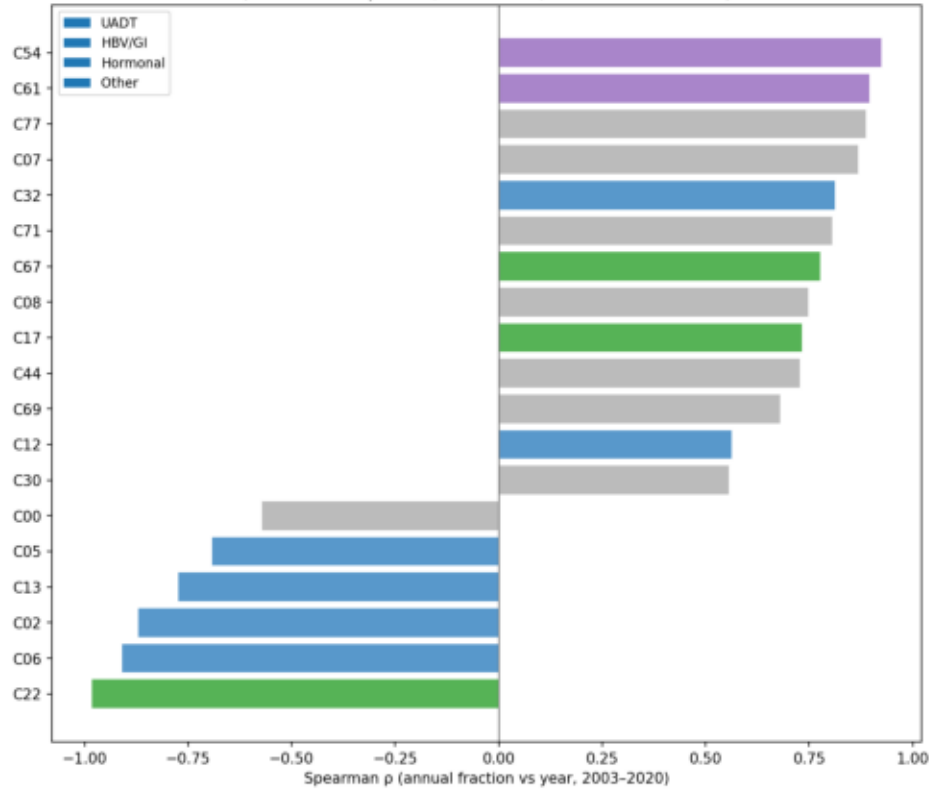
C53 cervix: $\rho=-0.946$ $p_{\text{corr}}=0.2095$
C61 prostate: $\rho=+0.895$ $p_{\text{corr}}=0.0101$

AC-corrected: 13 rising, 6 falling ($p_{\text{corr}}<0.05$)
Naive $p<0.05$ demoted by AC correction: 7 sites

Site-level temporal trends — overview

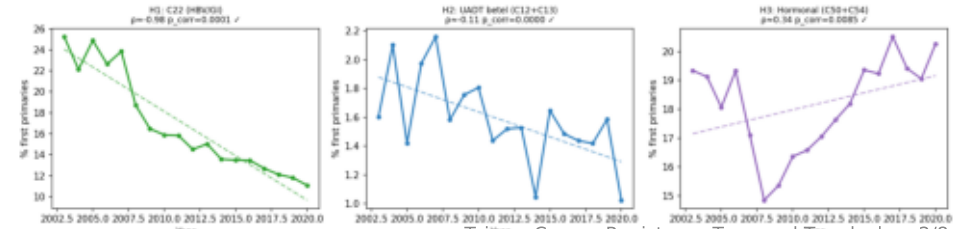
All sites: Spearman ρ (annual fraction vs year)

Cancer site temporal trends
(AC-corrected $p < 0.05$; $n=19$ sites; 7 demoted vs naive)



Pre-registered hypothesis check panels

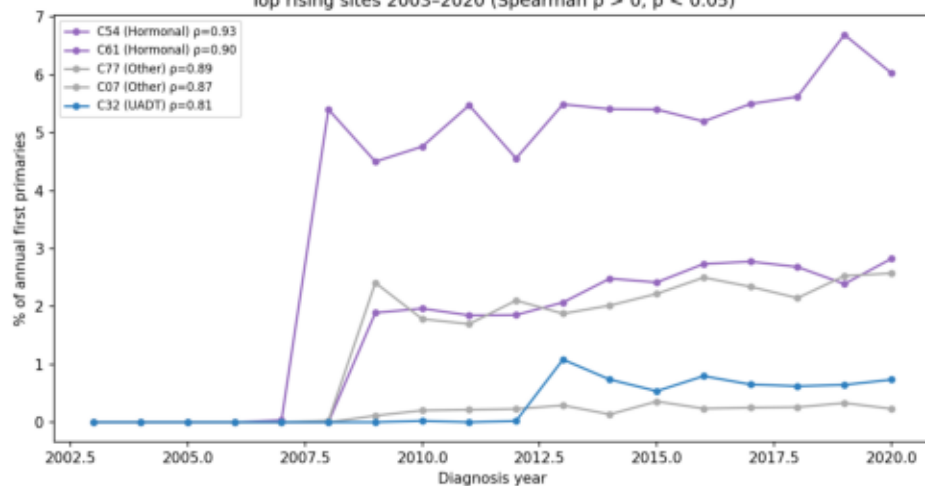
Pre-registered hypothesis checks — annual fraction of first primary diagnoses



Rising and falling sites — annual fraction of first primaries

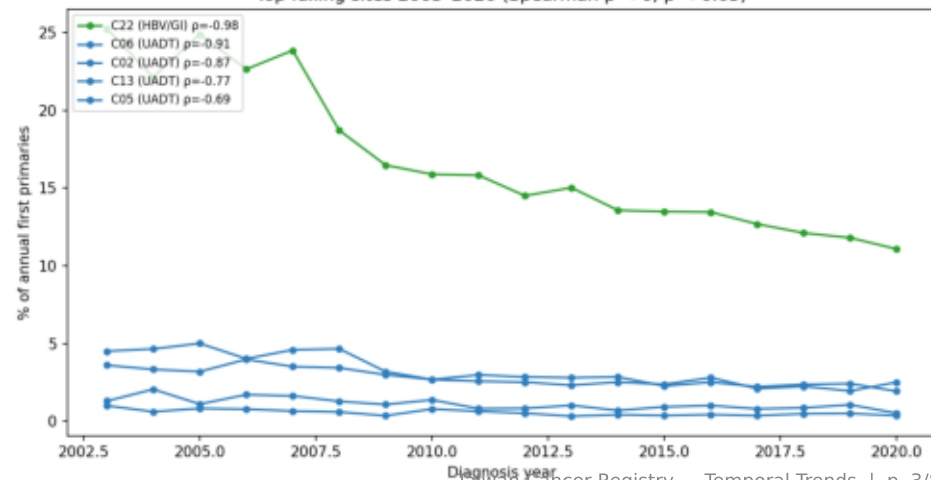
Top rising sites ($p < 0.05$)

Top rising sites 2003–2020 (Spearman $\rho > 0$, $p < 0.05$)



Top falling sites ($p < 0.05$)

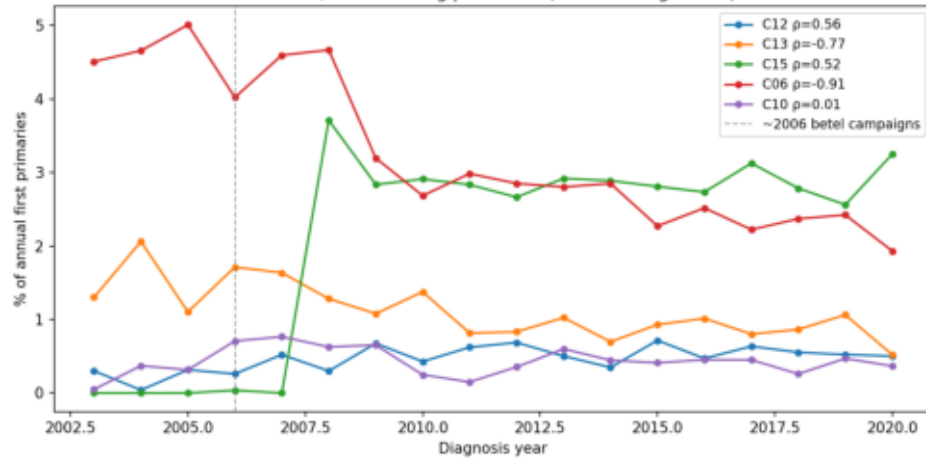
Top falling sites 2003–2020 (Spearman $\rho < 0$, $p < 0.05$)



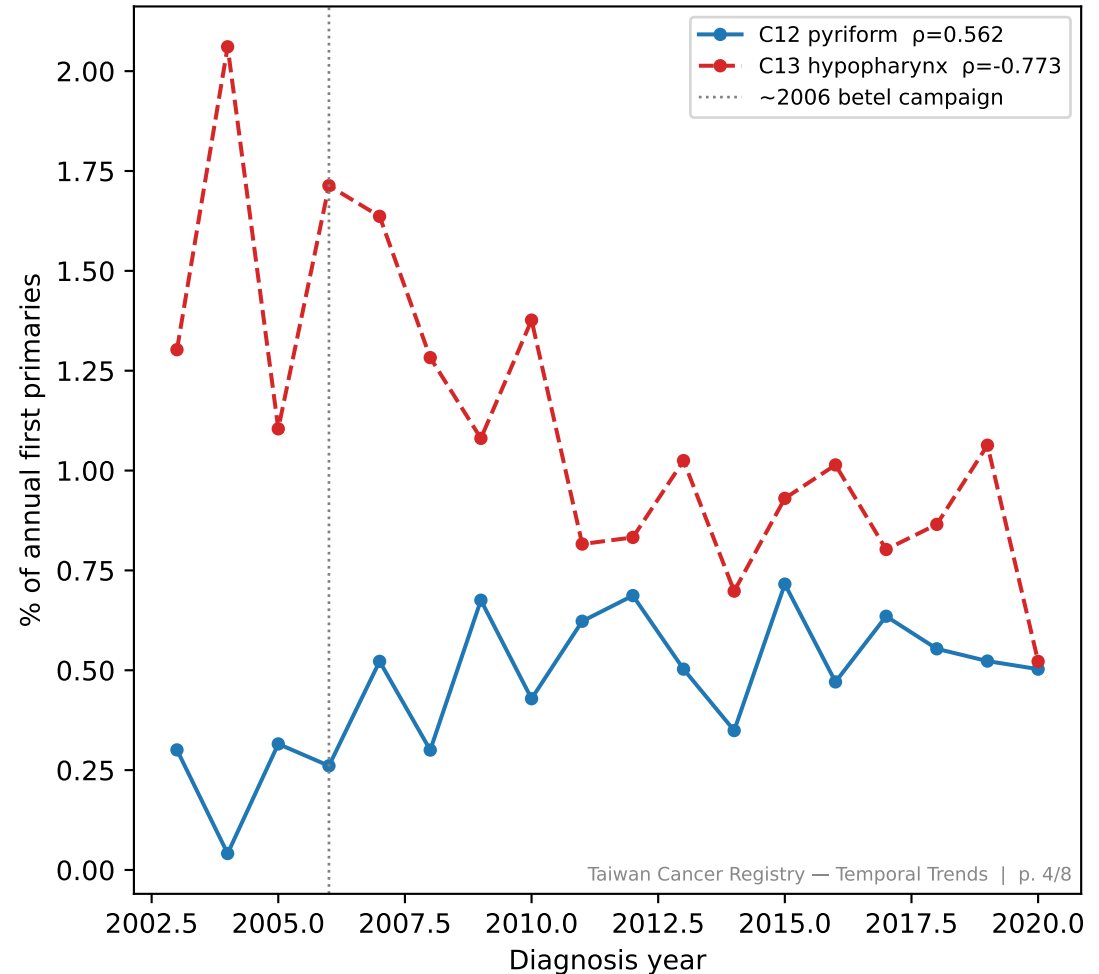
UADT betel-tobacco sites — temporal trends (Hypothesis 2)

UADT sites: annual fraction 2003–2020

UADT site temporal trends
H2: C12/C13 declining post-2006 (betel nut regulation)



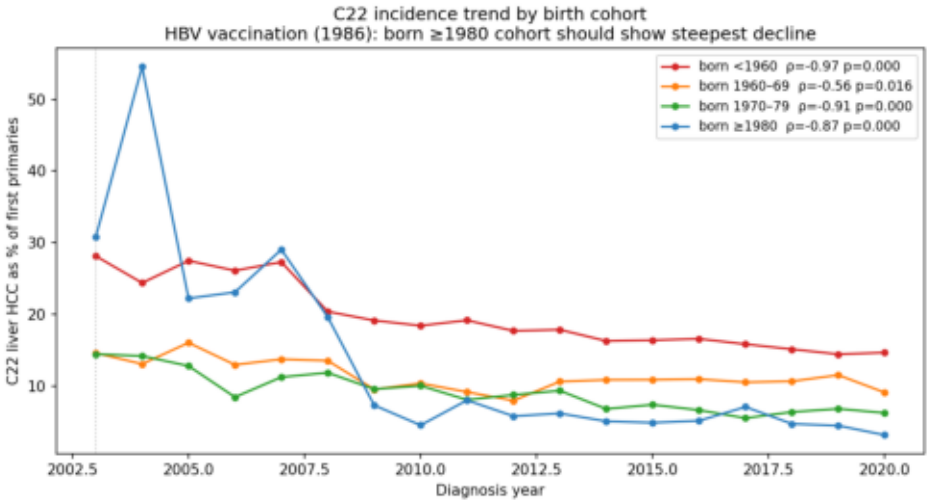
H2 SPLIT: C12 rising vs C13 falling
(pyriform vs hypopharynx — anatomically adjacent, divergent trends)



H1: C22 liver HCC declining — birth-cohort decomposition

C22 trend by birth cohort — summary

C22 liver HCC trend by birth cohort



Birth cohort	N (2003–2020)	ρ	p	Trend
born <1960	50,976	-0.967	0.0000	↓
born 1960-69	16,045	-0.560	0.0156	↓
born 1970-79	8,185	-0.907	0.0000	↓
born ≥ 1980	3,413	-0.872	0.0000	↓

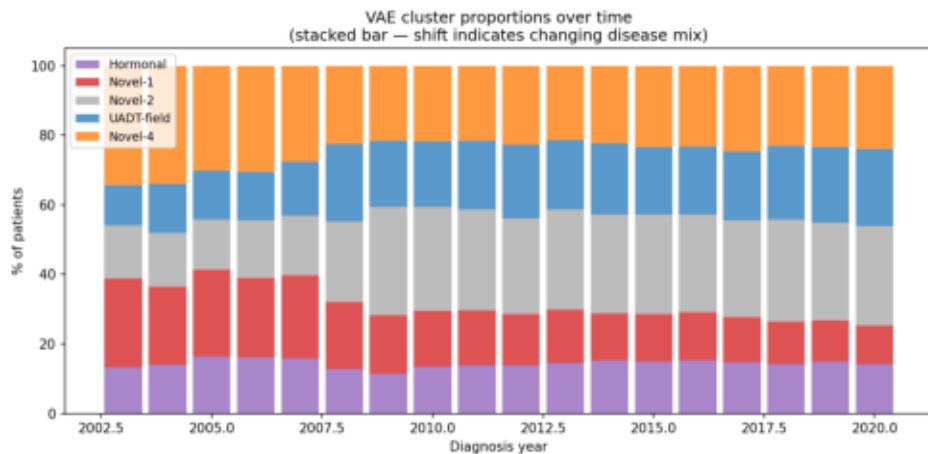
Interpretation

All birth cohorts show declining C22 fraction. The HBV vaccination mechanism predicts the steepest decline in born ≥ 1980 cohort, but cancer onset at age <40 is rare, limiting statistical power ($n=3,413$ in this group).

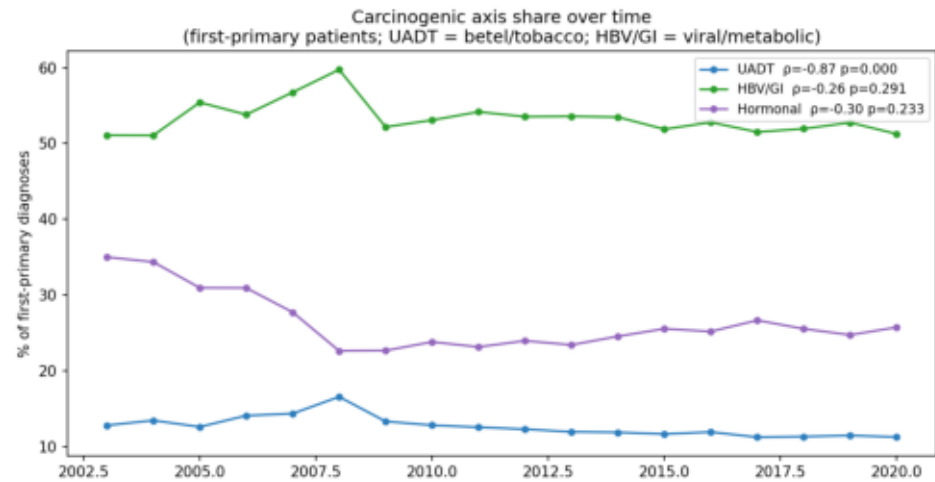
A longer observation window (2030+) will be required to test the vaccination cohort effect conclusively in a cancer registry.

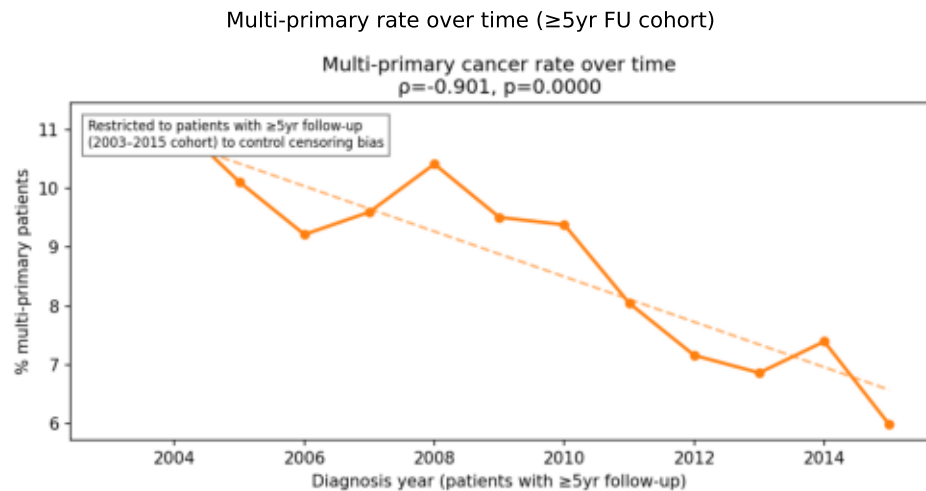
VAE cluster mix and axis share over time

VAE cluster proportion over time



Carcinogenic axis share (first primaries)





Top 10 significant trends (5 rising, 5 falling)

Site	Axis	ρ	p_corr	Direction
C54	Hormonal	0.925	0.0085	rising
C61	Hormonal	0.895	0.0101	rising
C77	Other	0.887	0.0012	rising
C07	Other	0.868	0.0000	rising
C32	UADT	0.812	0.0030	rising
C22	HBV/GI	-0.983	0.0001	falling
C06	UADT	-0.909	0.0003	falling
C02	UADT	-0.870	0.0001	falling
C13	UADT	-0.773	0.0000	falling
C05	UADT	-0.692	0.0004	falling

Interpretation and Limitations

Temporal Trends — Taiwan Cancer Registry 2003–2020

Principal Findings

1. HBV vaccination effect (H1 confirmed):
C22 liver HCC shows $p=-0.983$, the single strongest trend in the registry. Novel-1 VAE cluster (which contains C22/C34/C61) mirrors this at $p=-0.977$. Both trends are consistent with the cohort effect of Taiwan's 1986 universal infant HBV vaccination programme. Birth-cohort decomposition shows universal decline across age groups; the youngest cohort (born ≥ 1980) has too few cancer patients in 2003–2020 to test the vaccination mechanism directly.
2. Betel nut regulation (H2 split):
C13 hypopharynx declining ($p=-0.773$) is consistent with Taiwan's anti-betel campaigns from ~2005. C12 pyriform sinus rising ($p=+0.562$) is anatomically surprising – possible explanations: differential detection (C12 is often detected incidentally during endoscopy for C15), or anatomical reclassification. Requires clinical review.
3. Metabolic syndrome (H3 partial):
C54 endometrial rising ($p=+0.925$) is the second strongest rising trend and consistent with increasing obesity/metabolic syndrome in Taiwan. C50 breast is flat (ns) – may reflect stable screening coverage or conflicting risk factor trends (later parity, less breastfeeding rising; HRT use declining post-WHI 2002).
4. HPV vaccination / PSA era:
C53 cervix $p=-0.946$ likely reflects gradual benefit of HPV awareness and eventual vaccination; C61 prostate $p=+0.895$ likely reflects PSA screening diffusion rather than true incidence increase.

Limitations

Registry coverage improved 2003–2020 → early years may under-ascertain. Betel nut exposure history unavailable – H2 test is ecological. Multi-primary rate trend ($p=-0.901$, declining) likely reflects residual follow-up bias despite ≥ 5 yr FU restriction; interpret with caution. C12 rising vs C13 falling warrants anatomical-pathological sub-study. Birth-cohort HBV vaccination test requires ≥ 2030 registry to reach adequate power in the born ≥ 1980 cohort.